Capstone Project Phase 2

AI-Driven Image Processing for Organic Waste Classification and Biomethanation Potential Prediction in Bio-CNG Facilities

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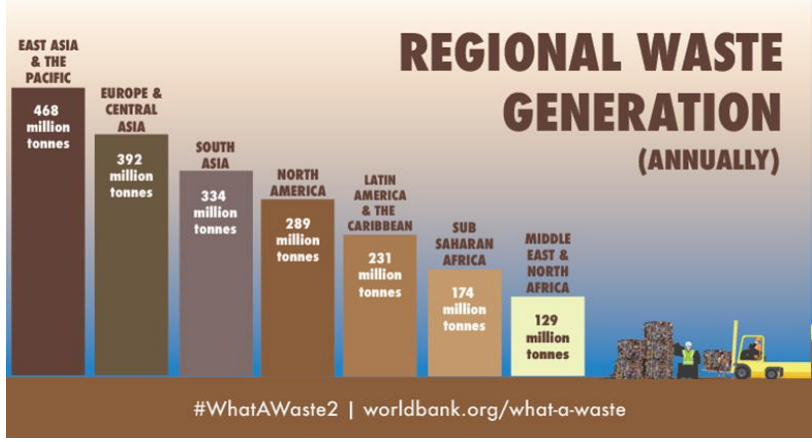
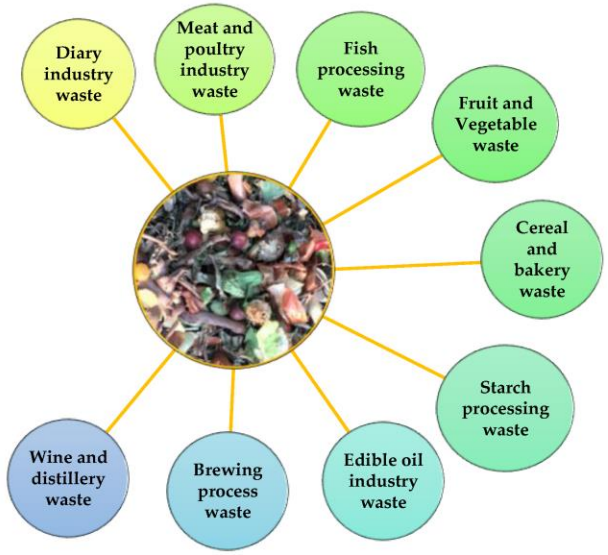
Introduction and Background

- Biodegradable wastes such as food waste and agricultural waste has a vast potential for renewable energy generation through the generation of biogas.
- Biogas plants typically rely on manual sorting and evaluation of organic waste, which is time-consuming and does not yield quick results.
- Recent advances in image processing facilitate the identification and classification of waste items based on visual inputs like images or video recordings.
- Use of visual waste identification coupled with existing Biochemical Methane Potential (BMP) values will improve the accuracy of the estimated energy yield.
- This project proposes a system that identifies biogas-potential waste in real-time and computes potential biogas yield, assisting efficient and sustainable plant operation.





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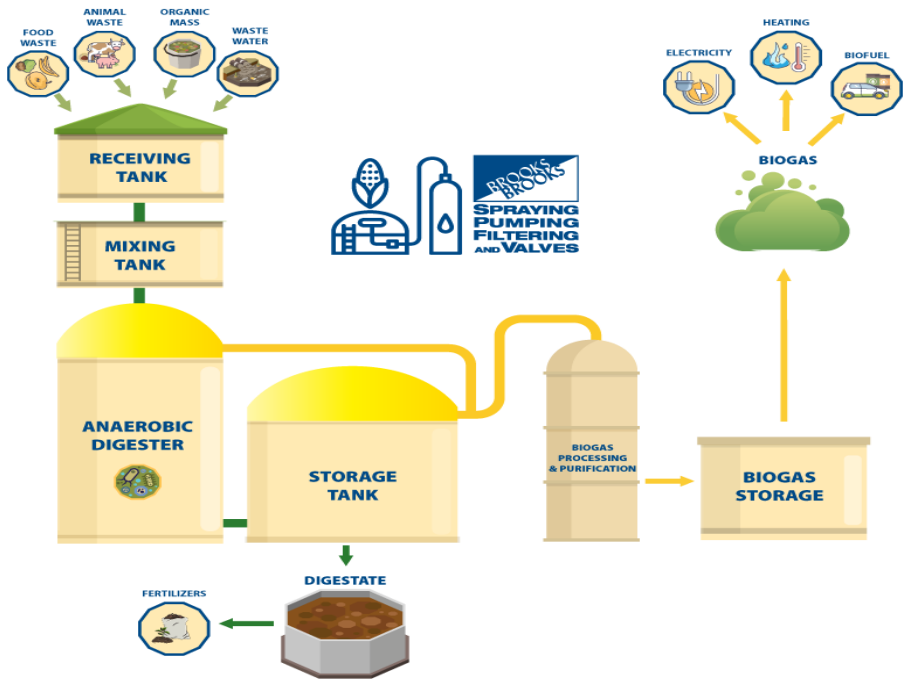
Objectives

- To develop a system capable of identifying organic waste with biogas potential using real-time images.
- To estimate the expected biogas yield by mapping detected waste types to their corresponding Biochemical Methane Potential (BMP) values.
- To provide an efficient, automated approach that aids in optimizing feedstock selection and improving the overall efficiency of biogas plant operations.



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Field Visit to SWMS Bio-Gas plant

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Challenges observed

- Waste types like banana peels, orange skins, and fiber rich leaves have low Biochemical Methane Potential (BMP) and high fiber content, slowing down digestion, increasing power and catalyst consumption.
- Lack of real-time assessment of biogas potential leads to inefficient feedstock mixing, negatively affecting biogas quality and overall yield.
- Organic waste is often mixed with non-biodegradable items like plastics, paper, and metals, requiring manual sorting which is time-consuming and labor-intensive.
- Sharp or solid objects (e.g., small iron pieces) choke or damage the crusher/debagger blades, causing frequent halts, increased downtime, and reduced plant efficiency.
- These challenges highlight the need for a system that can automatically identify biogas-suitable waste and predict output, enabling better planning, performance optimization, and cost reduction.

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Literature Review

TITLE	METHODOLOGY	RESEARCH GAP	YEAR
Modeling of biogas production from food, fruits and vegetables wastes using artificial neural network (ANN)	• Developed and evaluated multiple ANN architectures using a mixed-variable dataset from literature, achieving >85% prediction accuracy.	• Important biochemical parameters were excluded, data availability was limited, and the model lacks robustness and generalizability.	2020
The biomethane generation potential of wastes and wastewaters from the sericulture, fisheries, and agro-industrial sectors in India	• Estimated biomethane potential using waste generation rates, BMP values, and COD-based calculations for wastewaters.	• Lacks dynamic modeling, techno-economic and policy analysis, and suffers from regional data uncertainty.	2023
Towards Enhanced Prediction of Biomethane Production using Random Forest and Artificial Neural Network models	• Trained and tuned RF and ANN models on long-term multivariate data to predict biomethane production across time ranges.	• Limited model diversity, lack of interpretability, and insufficient analysis of hyperparameter influence.	2023

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Literature Review

TITLE	METHODOLOGY	RESEARCH GAP	YEAR
Innovative approach for predicting biogas production from large-scale anaerobic digester using long-short term memory (LSTM) coupled with genetic algorithm (GA)	• Applied LSTM with GA-based feature selection to predict biogas production under multiple digester input scenarios.	• Limited validation across diverse AD systems and absence of economic, environmental, and comparative model analysis.	2023
The Application of Artificial Intelligence in Waste Classification as an Effort In Plastic Waste Management	• Used transfer learning (EfficientNet-B0) with data augmentation for automated waste classification.	• Limited data diversity, category-specific underperformance, and lack of hybrid or adaptive models.	2024
Intelligent waste classification approach based on improved multi-layered convolutional neural Network	• Proposed an enhanced CNN with optimized depth, activation, and regularization, benchmarked against transfer learning models.	• Does not address continual learning, class imbalance robustness, or model interpretability.	2024
Machine learning based prediction of biogas generation from municipal solid waste: A data-driven approach	• Applied ML models (SVM, RF, XGBoost) with feature importance analysis to predict MSW-based biogas yield.	• Lacks biochemical process integration, validated key input parameters, and faces biomass composition constraints.	2024

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Methodology

Step 1: Dataset Collection

- **Classes:** ['banana_leaves', 'cabbage_leaves', 'orange_peels', 'plastics', 'rotten_bananas', 'rotten_lettuce', 'rotten_mangoes', 'rotten_potatoes', 'rotten_radish', 'rotten_tomatoes']
- **Dataset size:** 8212 images
- **Split dataset into train (70%), validation (15%), and test (15%) sets.**
- **Train images:** 5748
- **Validation images:** 1232
- **Test images:** 1232

Sample Images from Dataset

Step 2: Data Preprocessing

- Resize images to be 224 x 224
- Rotation : makes model invariant to small rotations.
- Width & Height Shift : simulates object translation within the frame.
- Shear : applies slanting/tilting transformations for geometric variety.
- Zoom : introduces scale variation, as if the camera zoomed in/out.
- Horizontal Flip: mirrors images horizontally to double viewpoint diversity.

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Methodology

Step 3: Model Design - Multiclass classification

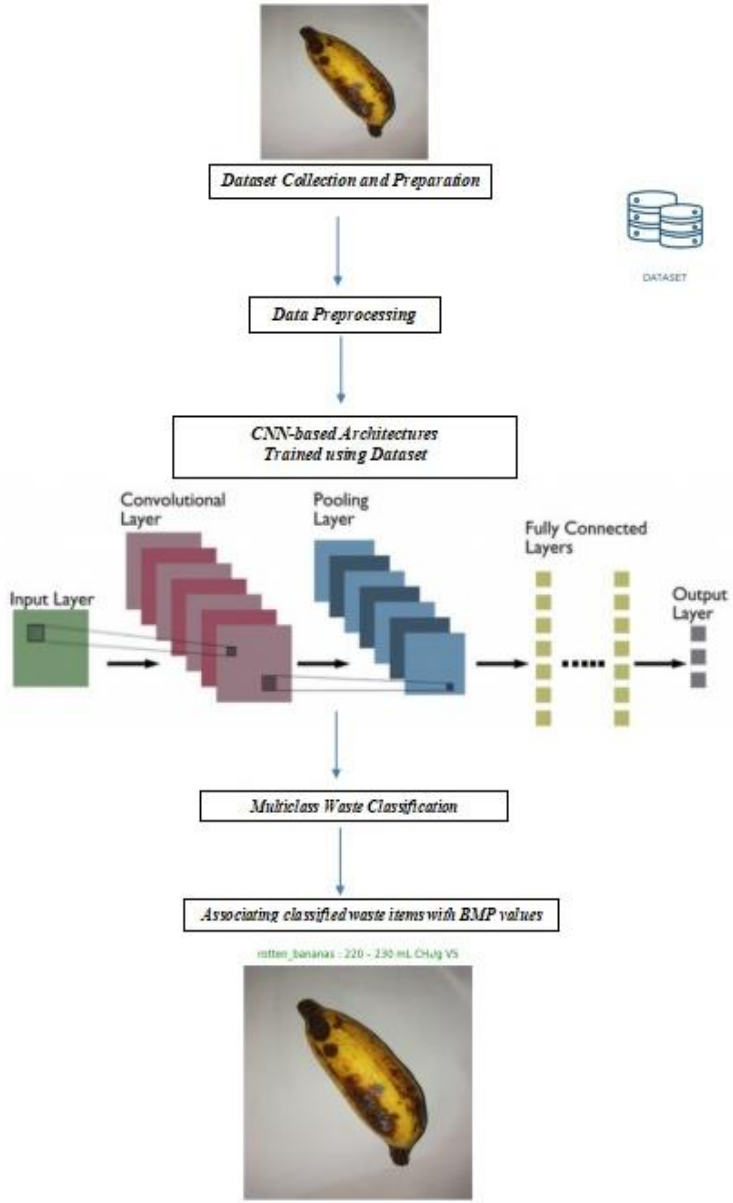
- Models Used: Trained five models — Vanilla CNN, VGG16, MobileNet, DenseNet121, EfficientNetB0
- Layers: Convolutional and pooling layers for feature extraction and dimensionality reduction
- Activation Functions: ReLU in hidden layers; Sigmoid and Softmax in output layer
- Regularization: Dropout layers to prevent overfitting
- Epochs & Training: Models trained for multiple epochs (10) with batch processing
- Transfer Learning: Pretrained weights used in VGG16, MobileNet, DenseNet121, EfficientNetB0
- Evaluation Metrics: Accuracy, F1-score, confusion matrix used for performance comparison
- Classified waste items are mapped with their existing BMP values.

Model Classification Output:

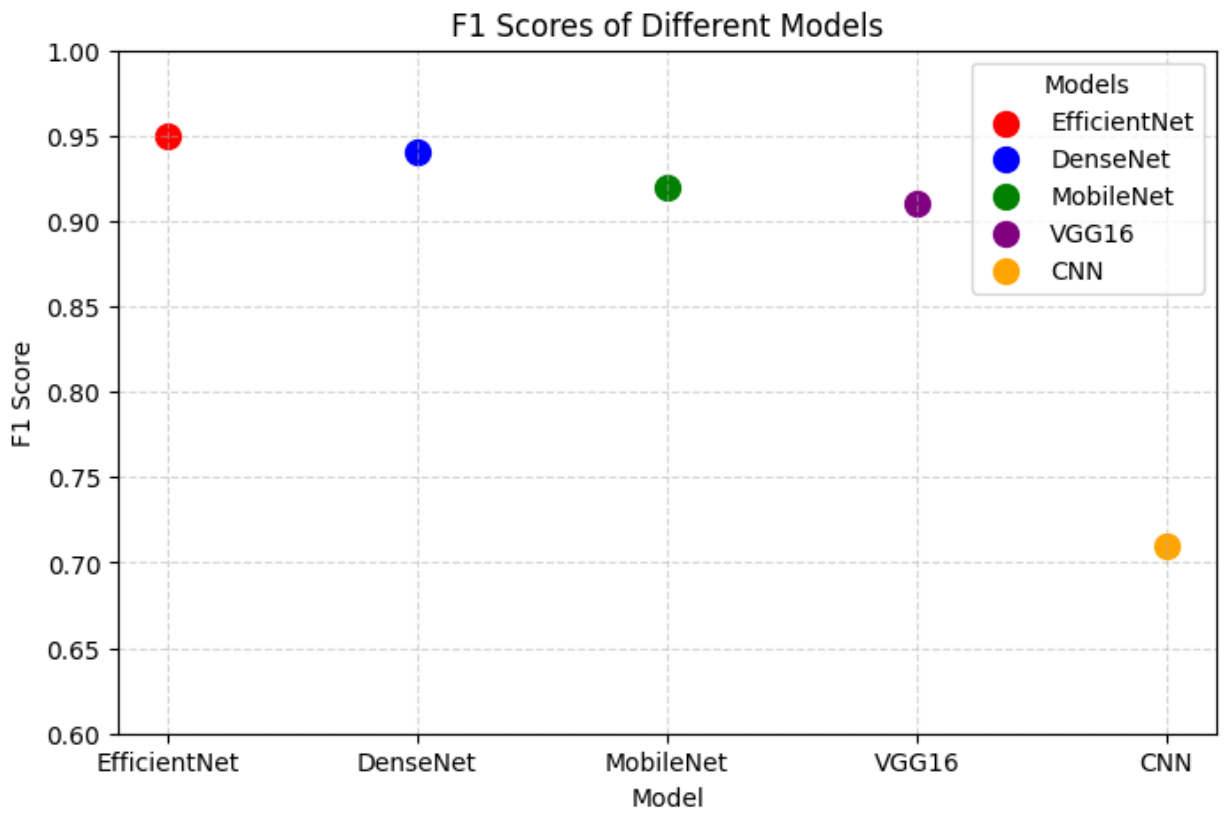
rotten_mangoes : 410 – 444 mL CH_4/g VS

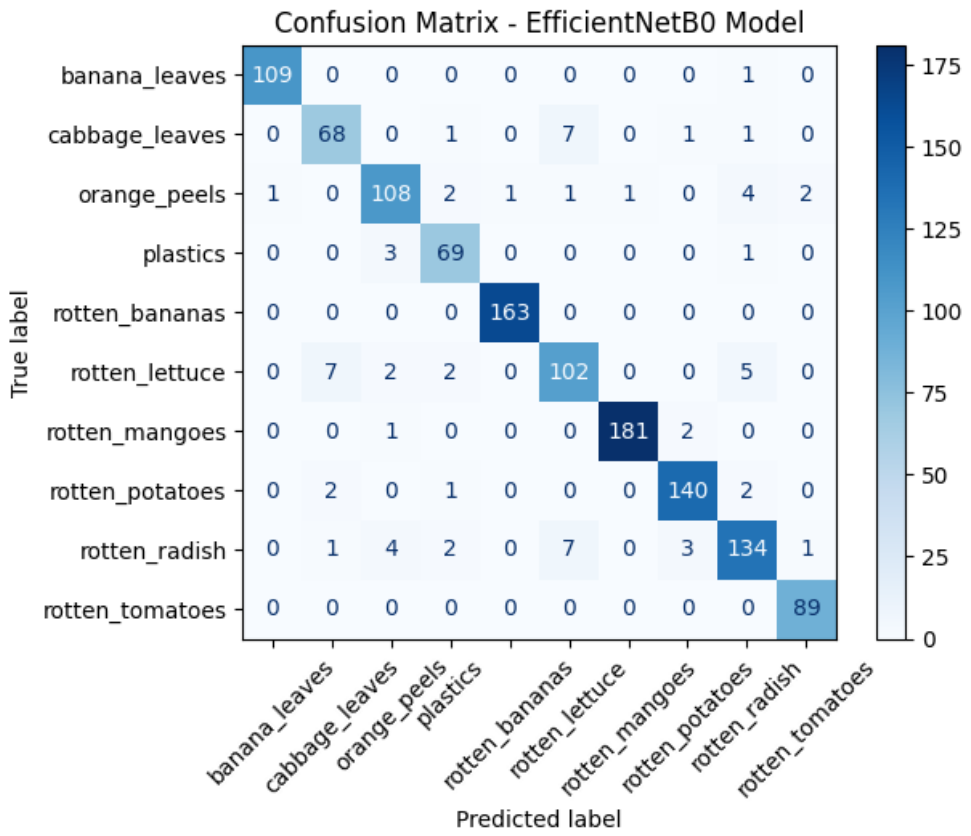
plastics : 0 mL CH_4/g VS

rotten_bananas : 220 – 230 mL CH_4/g VS



Results





Accuracy scores:

1)CNN: 0.74

2)VGG16: 0.91

3)MobileNet: 0.92

4)DenseNet: 0.93

5)EfficientNet: 0.94

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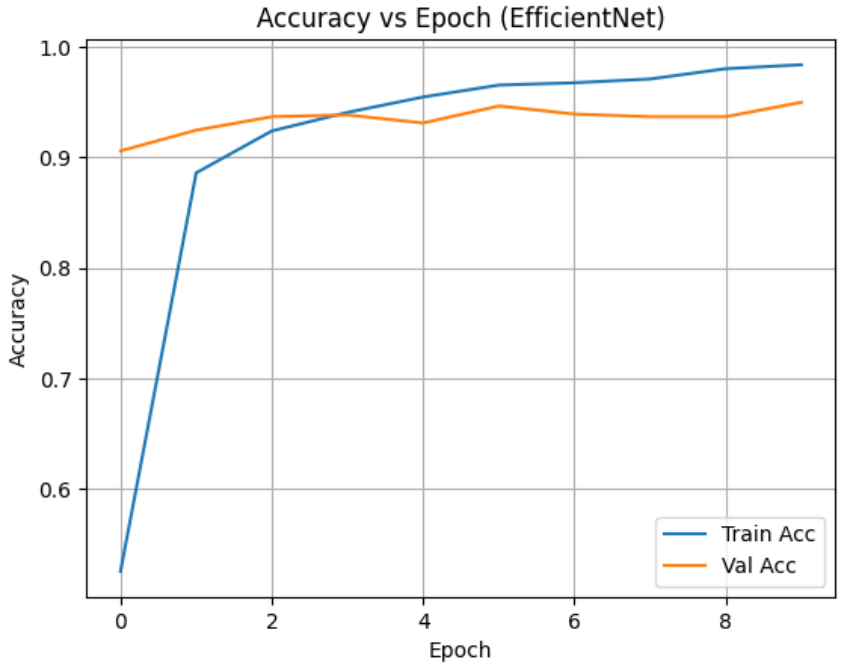
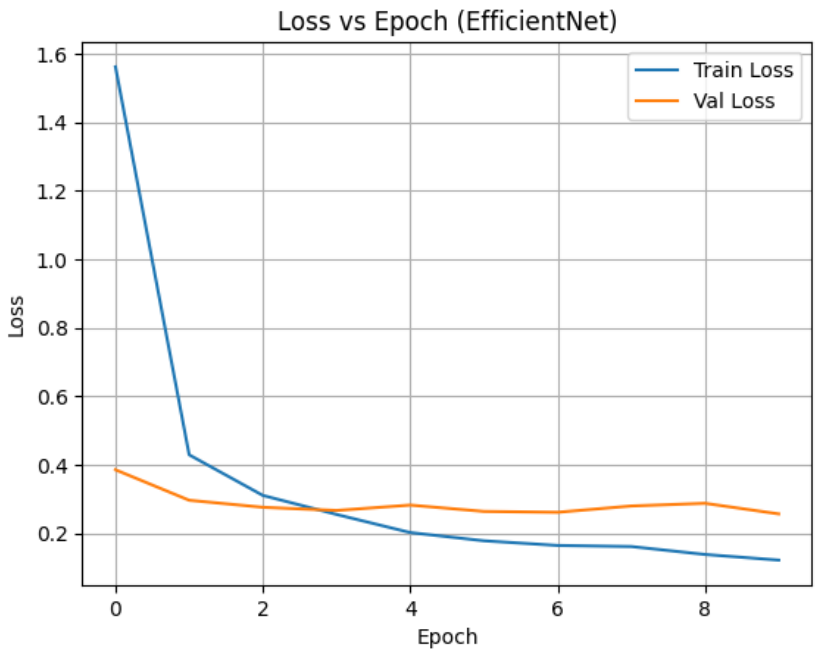
Results and Discussion

Accuracy and Loss curves:





- Accuracy curves show steady improvement and convergence, indicating strong learning and minimal overfitting.
- Loss curves decrease consistently, confirming stable and efficient model convergence.

- Among the evaluated models, EfficientNet achieved the highest accuracy (0.94), followed by DenseNet (0.93) and MobileNet (0.92), indicating their strong feature extraction and generalization capabilities for classification.
- The F1-score was also high for EfficientNet, confirming its superior balance between precision and recall.

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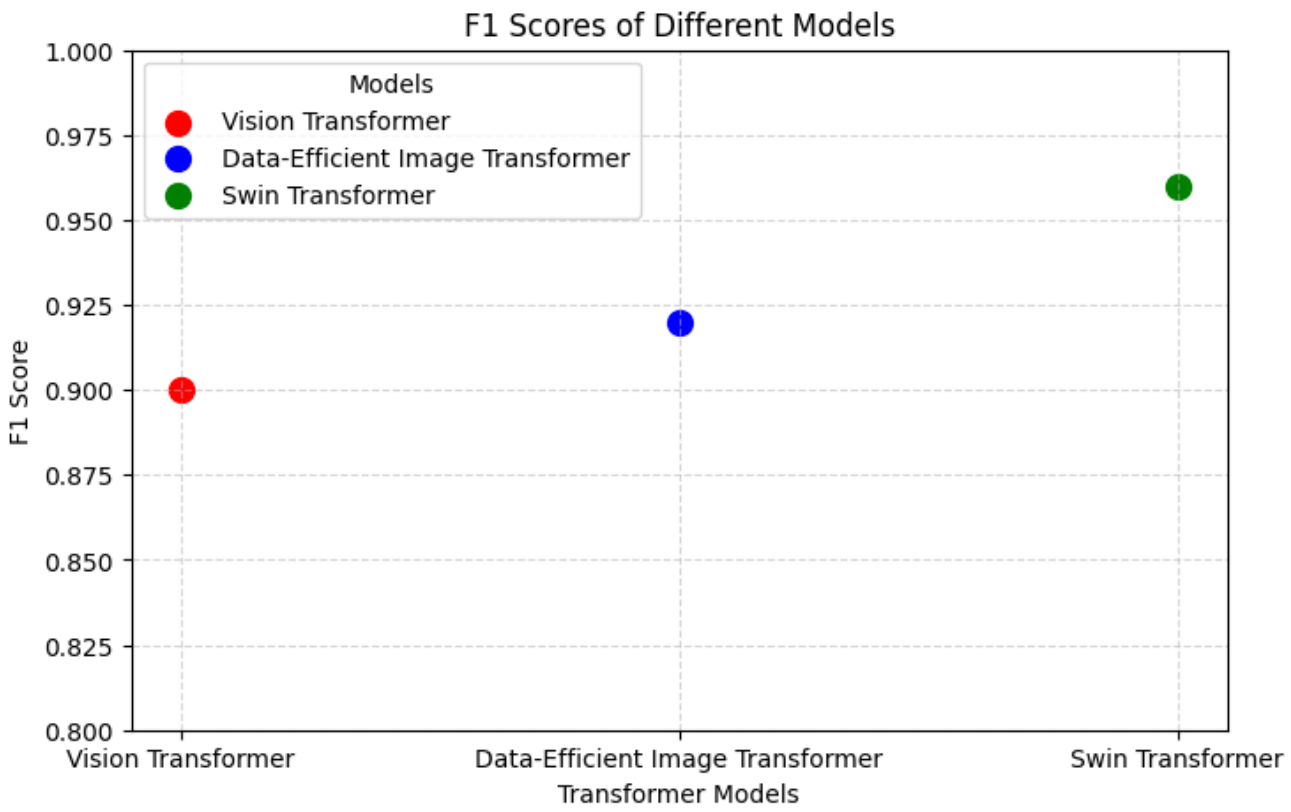
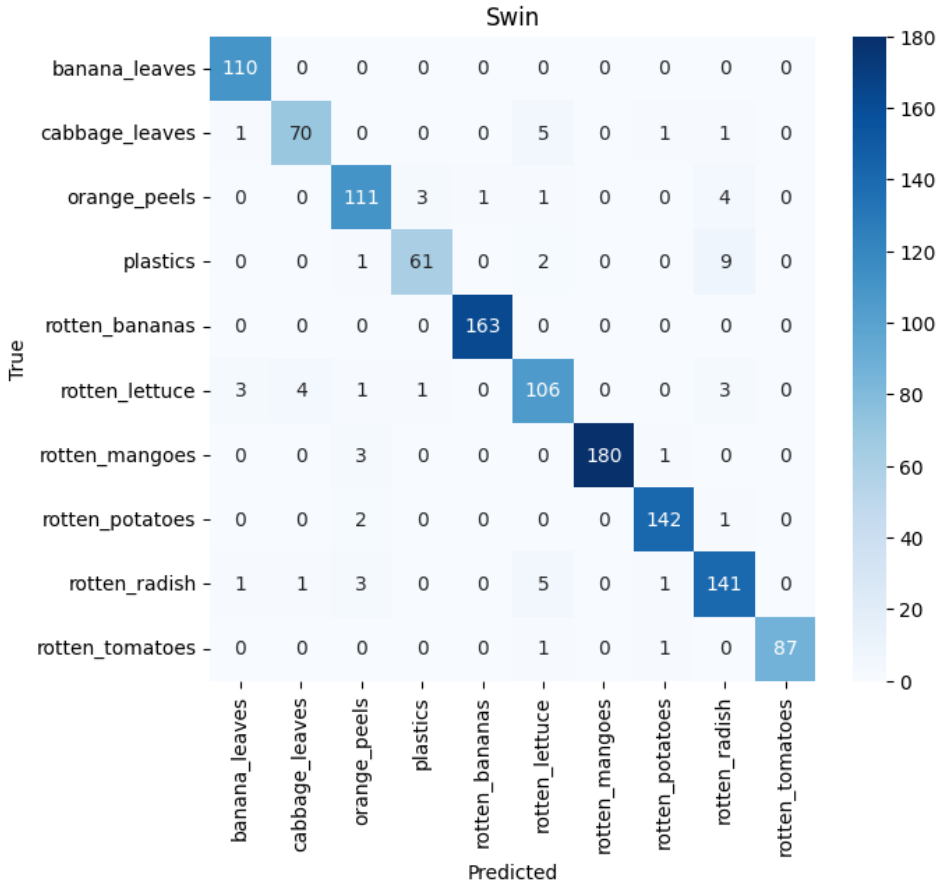
Results (Transformer Models)





Accuracy scores:

1)ViT : 0.92

2)DeiT : 0.93

3)Swin Transformer: 0.95

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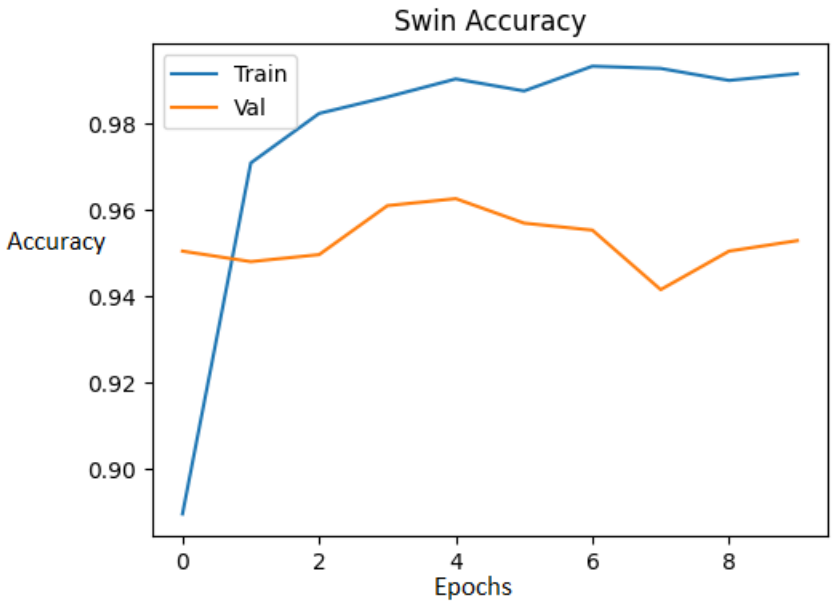
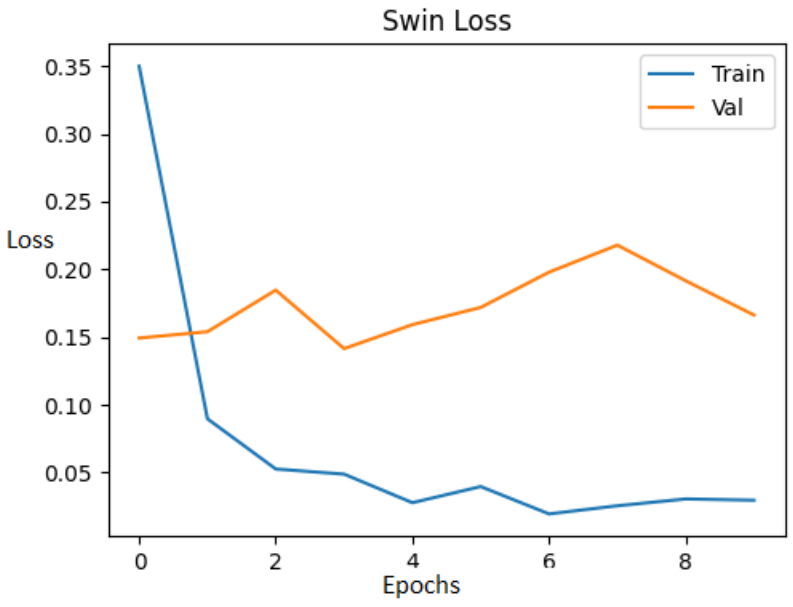
Results and Discussion

Accuracy and Loss curves:





- Accuracy curves show steady improvement and convergence, indicating strong learning and minimal overfitting.
- Loss curves decrease consistently, confirming stable and efficient model convergence.

- Among the evaluated models, achieved the highest accuracy by Swin (0.95), followed by DeiT (0.93) and ViT (0.92), indicating their strong feature extraction and generalization capabilities for classification.
- The F1-score was also high for Swin Transformer, confirming its superior balance between precision and recall.

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Expected Impacts and Conclusion

- Automated detection of biodegradable waste with biogas potential.
- Real-time estimation of potential biogas production.
- The system proposed offers an efficient means of real-time biogas-producing waste detection through image input with minimal human intervention.
- By assigning waste types to their respective BMP values, the system helps in the precise estimation of potential energy output.
- This technique enables improved planning and operation efficiency of biogas plants through automated feedstock evaluation.

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Future Work

- To incorporate visual multi-waste recognition, enabling simultaneous identification of multiple waste types within a single image.
- Develop a web-based application for system monitoring, data visualization, and user interaction.

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Dataset:

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Thank You